

DESIGN EVALUATION 2

XR Whiteboard Iteration 2 User Testing Plan

Name	Date	Prototype	Round
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What Round 1 Showed and How Iteration 2 Responds

Round 1 showed that the core classroom idea worked: all five participants completed the whiteboard and paper workflow and connected it with focused online learning or home study. The strongest surface was the wall whiteboard. The repeated weaknesses were the table tools, which all five participants initially found hard to notice or read; inconsistent paper drawing feedback; missed navigation; and difficulty seeing or looking between the desk and board. Iteration 2 therefore prioritises the seated lesson-and-note-taking experience before adding broader classroom features.

Table 1

Round 1 Findings and Implemented Responses

Round 1 finding	Evidence	Iteration 2 response to test
Table tools lacked visibility.	All five participants initially found the controls hard to notice or read.	The paper dock is larger, higher contrast, beside the paper, angled toward the seat, and limited to the nearest desk.
Paper drawing needed a focused retest.	P1 described slight glitches; P2 described the same surface as smooth.	Paper display and contact coordinates now share one orientation; stroke endings and fast endpoints are handled consistently.
Navigation and vertical view needed support.	P1 missed movement; P3 found looking up and down difficult.	Seat anchors place the viewer at the desk, and the seated view is designed to show the paper and teaching board together.
The board area needed a clearer interaction path.	Round 1 users relied on visible drawing controls and the board surface.	The unused black chalk object and front holder were removed; drawing controls remain beside the board.

Iteration 2 also adds session recording and playback, plus explicit simulator controller selection with T for the left controller and Y for the right controller. These are new capabilities rather than Round 1 findings, so they will be evaluated separately from the comparison measures.

Purpose and Research Questions

The purpose of this evaluation is to determine whether the revised prototype supports a complete classroom study workflow with minimal help. The test focuses on whether a seated learner can follow the teaching board, write personal notes, control both drawing surfaces, move without controller drift, and understand the new recording and simulator controls.

- Can participants see the teaching board and usable paper from the seated table view?
- Can participants discover the paper tools and produce reliable notes without direct instruction?
- Does navigation move the learner and controllers together at a comfortable, predictable rate?
- Can participants understand and complete Record, Stop, and Play on the whiteboard?
- In the desktop simulator, do T and Y clearly activate the intended left and right controllers?

Testing Aim and Success Measures

The main aim is to verify that the changes derived from Round 1 remove the repeated barriers while preserving the successful two-surface classroom workflow. Results will be reported by participant rather than inferred from technical checks.

Table 2*Success Measures for Iteration 2*

Measure	Success target	Evidence to collect
Shared seated view	At least 4 of 5 participants can see the full paper and relevant board content from the seated pose without moving the chair or receiving help.	First seated view, head movement, repositioning, and participant rating.
Paper tool discovery	At least 4 of 5 open TOOLS and select Pencil within 15 seconds without a direct hint.	Time to first correct selection and assistance level.
Paper drawing	At least 90% of required lines and circles appear in the intended position without gaps, axis reversal, or unwanted continuation.	Completed stroke count, visible defects, and input method.
Lesson-to-notes workflow	At least 4 of 5 copy three items from the board to paper while remaining seated and report that switching attention is comfortable (4 or 5 out of 5).	Task completion, gaze or head movement, comfort rating, and comments.
Navigation stability	All observed movement keeps both controllers aligned with the learner; at least 4 of 5 rate movement predictability at 4 or 5.	Relative hand position, overshoot, correction, and rating.
Whiteboard recording	At least 4 of 5 complete Record–draw–Stop–Play and recognise that playback reproduces board actions without audio.	Completion, errors, replay match, and explanation in the participant's own words.
Simulator controller choice	In the simulator check, T moves only the left controller, Y moves only the right controller, and Escape returns to cursor mode in every attempt.	Moderator pre-test and participant task log.
Overall confidence	Median confidence is at least 4 out of 5, with no unresolved severe comfort or blocking issue.	Post-test rating and interview response.

Method**Participants**

Recruit five university students who represent the intended study context. Aim for at least three participants who did not take part in Round 1 so that discoverability is measured without prior learning. Up to two returning participants may be included to compare whether the revised view and controls address their earlier difficulty. Record prior VR experience as none, occasional, or frequent.

Equipment and Build

- Use the same Iteration 2 build, blank paper, whiteboard state, table destination, and task order for every participant.
- Prefer a Meta Quest 2 or Quest 3 for comfort, reach, controller tracking, and performance evidence. Record the headset model and build identifier.
- Run the T/Y hand-selection task in the Unity XR Device Simulator because T and Y are keyboard simulator controls. On a headset, verify both physical controllers instead and mark T/Y as not applicable.
- Prepare a timer, observation sheet, consent record, and optional screen recording. Do not record audio or video without explicit participant consent.

Session Length and Safety

Allow approximately 20 minutes: 3 minutes for briefing and calibration, 12 minutes for tasks, and 5 minutes for questions. Give the participant a clear play area and seated position. Tell them they may pause or stop at any time. Stop immediately if they report nausea, dizziness, eye strain, loss of tracking, or discomfort. Store results under participant identifiers P01–P05 and keep consent details separate from observations.

Facilitation and Assistance

Use think-aloud moderation and read every task instruction exactly. Wait before helping. If the participant is stuck, use a neutral prompt first, such as “What options can you see?” Record assistance as none, neutral prompt, directional hint, or direct help. The facilitator should not point to a control before discoverability time has been recorded.

Test Setup and Task Sequence

Table 3

Standardised Task Sequence

Task	Instruction read to participant	Observe and record
0. Pre-test check	Moderator only: open the correct build, clear both surfaces, confirm tracking, movement, and the controller mode required for this session.	Build/device ID, errors, frame or tracking issue, T/Y pre-check if using the simulator.
1. Explore	Look around and tell me what you think you can do in this classroom.	Whether the participant identifies the board, paper, navigation, and study purpose without prompting.
2. Sit at a table	Move to Table 1 and sit in the study position. Tell me what you can see without leaving the seat.	Navigation discovery; final head pose; whether paper, tools, and teaching board are visible; overshoot or correction.
3. Follow and take notes	From your seat, copy the date, a horizontal line, and a circle shown on the teaching board onto your paper.	Attention changes between board and paper, readability, view comfort, paper location and orientation.
4. Paper tools	Use the paper controls to select Pencil, draw a vertical line, erase part of it, and clear only the paper.	Time to find TOOLS, first selection, stroke continuity, eraser precision, clear scope, assistance.
5. Move and return	Move to another table, then return to Table 1. Describe how the movement and controller speed feel.	Camera/controller alignment, unexpected hand drift, speed, destination accuracy, comfort.
6. Whiteboard	Go to the whiteboard. Draw a black horizontal line, red vertical line, blue circle, and green L. Erase part of one shape, then clear the board.	Tool selection, colour mapping, stroke quality, clear confirmation, obstruction or confusion from removed props.
7. Record and replay	Select Record, draw two shapes in different colours, pause briefly, select Stop, then select Play. Explain what you think was replayed.	Control discovery, disabled states, action order, colour, pause, final result, expectation of audio.
8. Choose controllers	Simulator only: press T and aim the left controller. Then press Y and aim the right controller. Press Escape when finished.	Only the selected controller moves and triggers; camera is stable; participant understands active-hand state.
9. Final reflection	Complete the ratings and answer the interview questions.	Confidence, comfort, perceived usefulness, repeated barriers, and improvement priority.

Note. Before Task 3, place the date, horizontal line, and circle on the teaching board. Use identical content for all participants. This task evaluates the combined view rather than memory or handwriting quality.

Questions Used

- Before testing: What do you expect to do in this classroom, and what do you think the paper is for?
- Could you see the teaching board and paper together from the seated view? What did you have to adjust?
- How did drawing on the revised paper compare with drawing on the wall whiteboard?
- Were the paper tools easy to notice, read, and reach? Which control was least clear?
- Did the controllers stay with you while moving, or did either controller seem faster than your movement?
- What did T and Y do? Which controller felt active after each key? (Simulator sessions only.)

- What did Record, Stop, and Play capture? Did the replay match what you expected?
- Rate view comfort, movement predictability, paper reliability, control clarity, and overall confidence from 1 to 5.
- What was the easiest part, the most difficult part, and the single most important improvement?

Data Collection Sheet

Complete one copy per participant. Use factual observations before interpreting the cause of a problem.

Table 4

Participant and Session Record

Field	Record
Participant ID	P_____
Date and facilitator	_____
New or returning participant	☐ New ☐ Returning from Round 1
Prior VR experience	☐ None ☐ Occasional ☐ Frequent
Device and input	☐ Quest 2 ☐ Quest 3 ☐ Simulator Other: _____
Build identifier	_____
Recording consent	☐ Notes only ☐ Screen ☐ Audio ☐ Declined recording
Session outcome	☐ Completed ☐ Paused ☐ Stopped; reason: _____

Table 5

Task Observation Record

Task	Result and assistance	Time/attempts	Observed issue or quote
Explore	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
Seat and shared view	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
Follow and take notes	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
Paper tools and strokes	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
Move and return	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
Whiteboard workflow	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
Record and replay	☐ Pass ☐ Prompt ☐ Help ☐ Fail	_____	_____
T/Y controller choice	☐ Pass ☐ Prompt ☐ Help ☐ N/A	_____	_____

Table 6*Post-Test Ratings*

Rating (1 = very poor; 5 = excellent)	1	2	3	4	5	Comment
Seated view comfort	[]	[]	[]	[]	[]	_____
Paper drawing reliability	[]	[]	[]	[]	[]	_____
Paper control clarity	[]	[]	[]	[]	[]	_____
Movement predictability	[]	[]	[]	[]	[]	_____
Recording control clarity	[]	[]	[]	[]	[]	_____
Overall confidence	[]	[]	[]	[]	[]	_____

Analysis Plan**Quantitative Analysis**

- Report unassisted completion, prompted completion, direct-help completion, and failure for each task.
- Calculate the median rating and the number of participants rating each measure 4 or 5. Do not report percentages without the participant count because the sample is five.
- For paper reliability, divide acceptable required strokes by all required strokes and describe each visible defect.
- Compare tool discovery, paper reliability, and view comfort with the Round 1 evidence. Keep returning and new participant observations identifiable in the analysis.

Qualitative Analysis

Group factual observations and participant comments under shared view, tool discoverability, paper drawing, movement and controller alignment, recording, comfort, and perceived learning value. Distinguish a repeated pattern from a single preference. Use short participant quotations only when recording consent permits it, and label them with participant identifiers.

Decision Rules**Table 7***Interpretation and Iteration Rules*

Decision	Rule	Next action
Validated	The success target is met and no severe comfort, tracking, or data-loss issue occurs.	Keep the design and document minor refinements.
Validated with issue	The target is met, but one participant encounters a repeatable non-blocking problem.	Retest the exact trigger and make a bounded correction.
Revise	Two or more participants encounter the same barrier, or fewer than 4 of 5 complete a core task without direct help.	Prioritise the repeated barrier for the next prototype and repeat that task.
Technical blocker	The build crashes, tracking fails, controller movement separates from the learner, drawing data is lost, or replay is materially incorrect.	Stop the affected task, preserve evidence, repair, and rerun the session on a verified build.

Moderator Checklist and Evidence Record

Table 8

Before, During, and After Each Session

Stage	Checklist
Before	☐ Consent recorded ☐ Safe area checked ☐ Correct build/device logged ☐ Surfaces cleared ☐ Input/tracking checked ☐ Board prompt prepared
During	☐ Script read consistently ☐ Think aloud encouraged ☐ Discoverability time captured ☐ Every hint coded ☐ Comfort monitored ☐ Defects described factually
After	☐ Ratings complete ☐ Final questions asked ☐ Evidence labelled by participant ID ☐ Consent limits applied ☐ Build/device noted ☐ Data stored securely

The Iteration 2 report should state the actual participant count, device, incomplete sessions, assistance, and evidence limits. If Quest testing is unavailable, report simulator results as simulator evidence and keep headset comfort, tracking, reach, and performance as unresolved. Do not convert the moderator's existing technical validation into participant results.

Planned Output

The completed evaluation will include a participant result table, factual observations, themes, a check against every success measure, design responses, and a short reflection on method. Supporting evidence will consist of consent-compatible notes, task sheets, build information, and recordings where permission was granted. The next design decision will be based on repeated participant evidence and the decision rules above.